

SIM70X0(MDM9205) DAM Quick Start

CONTENTS



01
DAM Introduction



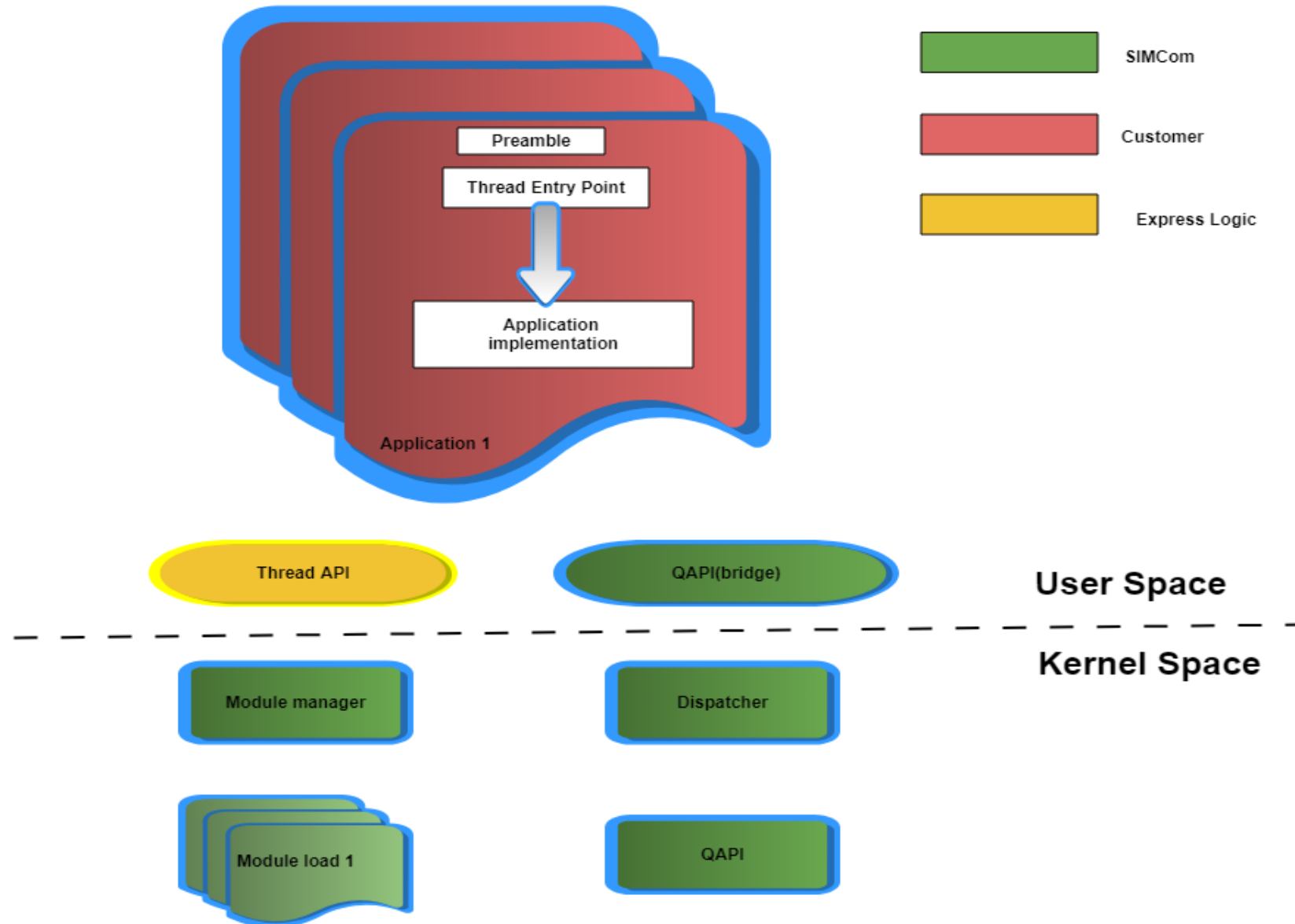
02
DAM SDK Compile



03
DAM Download and Run



04
Perspectives



Why Downloadable Application Module

- **Powerful:** based on C language, which is powerful programming language ,DAM can perform complex calculation as programming on module firmware directly, and run in ARM Cortex-A7(up to 1.3 GHz with 256 KB L2 cache).
- **Simplicity:** provides many interfaces to assist customer develop applications easily including AT command operation, thread operation, EFS operation, GPIO/I2C/ADC and other operation libraries.
- **Efficiency:** C language is the fastest language besides ASM language. By using it, customers do not need to care whether their application can perform so quickly to handle software/hardware operation .
- **Extensibility:** provides many easy-to-use interface which can provide most commonly-used functions. Customer can also develop their own libraries like database(sqlite) or compress utility(zip) easily.

Why Downloadable Application Module

- **Multi-threads support:** supports multi thread running in the module, by using the thread APIs, customer's applications can create many powerful functions.
- **Big memory:** provides up to 3MB DDR memory, 500KB code backup flash and about 5MB EFS space for customer applications to use. With big memory support, customer can write very complicated applications with multiple C source files.
- **Powerful periphery operation:** provides powerful APIs to let customer applications operate numerous periphery devices.
- **Flexible:** don't rely on FW version of module, DAM could run successfully even after the FW version has been updated!
- **Good power utilizing:** allows to save more power than using external MCU. Also, when DAM is in idle mode (call tx_event_flags_get or tx_thread_sleep), the module will enter sleep mode as it needs. Any interrupt will wakeup the module

SIM7070G/SIM7070E HW Peripheral



Pin Name	Pin No.	Mode 1	Mode 2	Mode 3	Mode 4	Interrupt	Remark
UART_RI	4	GPIO26	-	-	-	Y	
UART_DCD	5	GPIO28	-	-	-	Y	
PCM_CLK	11	GPIO24	-	-	-	N	
PCM_SYNC	12	GPIO21	-	-	-	Y	
PCM_DIN	13	GPIO22	-	-	-	Y	
PCM_DOUT	14	GPIO23	-	-	-	N	
GPIO1	19	GPIO4	SPI2_MOSI	UART2_TX	-	Y	boot_config[3]
GPIO2	20	GPIO5	SPI2_MISO	UART2_RX	-	Y	
GPIO3	21	GPIO7	SPI2_CLK	UART2_RT S	I2C2_SCL	N	
UART_RXD1	22	GPIO1	UART1_RX	SPI1_MISO	-	Y	QAPI_UART_PORT_001_E
UART_TXD1	23	GPIO0	UART1_TX	SPI1_MOSI	-	Y	QAPI_UART_PORT_001_E
ADC	25	ADC	-	-	-	N	

SIM7070G/SIM7070E HW Peripheral



Pin Name	Pin No.	Mode 1	Mode 2	Mode 3	Mode 4	Interrupt	Remark
I2C_SDA	37	GPIO2	UART1_CTS	I2C1_SDA	SPI1_CS	Y	
I2C_SCL	38	GPIO3	UART1_RTS	I2C1_SCL	SPI1_CLK	N	
GPIO4	48	GPIO6	SPI2_CS	UART2_CTS	I2C2_SDA	Y	
UART_RXD3	49	GPIO9	SPI3_MISO	UART3_RX	-	Y	QAPI_UART_P ORT_002_E
UART_TXD3	50	GPIO8	SPI3_MOSI	UART3_TX	-	Y	QAPI_UART_P ORT_002_E
NETLIGHT	52	GPIO11	SPI3_CLK	I2C3_SCL	UART3_RTS	N	
STATUS	66	GPIO10	SPI3_CS	I2C3_SDA	UART3_CTS	N	
GPIO5	67	GPIO30	-	-	-	Y	
GPIO6	68	GPIO52	-	-	-	Y	

SIM7080G HW Peripheral



Pin Name	Pin No.	Mode 1	Mode 2	Mode 3	Mode 4	Interrupt	Remark
UART_DCD	5	GPIO28	-	-	-	Y	
UART_RI	7	GPIO26	-	-	-	Y	
PCM_DIN	9	GPIO22	-	-	-	Y	
PCM_DOUT	10	GPIO23	-	-	-	N	
PCM_CLK	11	GPIO24	-	-	-	N	
PCM_SYNC	12	GPIO21	-	-	-	Y	
UART_TXD1	22	GPIO0	UART1_TX	SPI1_MOSI	-	Y	QAPI_UART_P ORT_001_E
UART_RXD1	23	GPIO1	UART1_RX	SPI1_MISO	-	Y	QAPI_UART_P ORT_001_E
ADC	38	ADC	-	-	-	N	
NETLIGHT	41	GPIO11	SPI3_CLK	I2C3_SCL	UART3_RTS	N	
STATUS	42	GPIO10	SPI3_CS	I2C3_SDA	UART3_CTS	N	

SIM7080G HW Peripheral



Pin Name	Pin No.	Mode 1	Mode 2	Mode 3	Mode 4	Interrupt	Remark
SPI_CS	48	GPIO6	SPI2_CS	UART2_CTS	I2C2_SDA	Y	
SPI_MOSI	49	GPIO4	SPI2_MOSI	UART2_TX	-	Y	boot_config[3]
SPI_CLK	50	GPIO7	SPI2_CLK	UART2_RTS	I2C2_SCL	N	
SPI_MISO	51	GPIO5	SPI2_MISO	UART2_RX	-	Y	
GPIO1	57	GPIO50	-	-	-	Y	
GPIO2	58	GPIO30	-	-	-	Y	
GPIO3	59	GPIO25	-	-	-	N	
GPIO4	60	GPIO52	-	-	-	Y	
UART_TXD3	61	GPIO8	SPI3_MOSI	UART3_TX	-	Y	QAPI_UART_PORT_002_E
UART_RXD3	62	GPIO9	SPI3_MISO	UART3_RX	-	Y	QAPI_UART_PORT_002_E
I2C_SDA	64	GPIO2	UART1_CTS	I2C1_SDA	SPI1_CS	Y	
I2C_SCL	65	GPIO3	UART1_RTS	I2C1_SCL	SPI1_CLK	N	

SIM7090G HW Peripheral



Pin Name	Pin No.	Mode 1	Mode 2	Mode 3	Mode 4	Interrupt	Remark
GPIO1	1	GPIO50	-	-	-	Y	
PCM_DIN	2	GPIO22	-	-	-	Y	
PCM_CLK	3	GPIO24	-	-	-	N	
UART_RXD3	4	GPIO9	SPI3_MISO	UART3_RX	-	Y	QAPI_UART_PORT_002_E
I2C_SDA	5	GPIO2	UART1_CTS	I2C1_SDA	SPI1_CS	Y	
SPI_MISO	8	GPIO5	SPI2_MISO	UART2_RX	-	Y	
SPI_CLK	9	GPIO7	SPI2_CLK	UART2_RTS	I2C2_SCL	N	
ADC0	17	ADC	-	-	-	-	
ADC1	18	ADC	-	-	-	-	
PCM_DOUT	34	GPIO23	-	-	-	N	
PCM_SYNC	35	GPIO21	-	-	-	Y	
UART_TXD3	36	GPIO8	SPI3_MOSI	UART3_TX	-	Y	QAPI_UART_PORT_002_E
I2C_SCL	37	GPIO3	UART1_RTS	I2C1_SCL	SPI1_CLK	N	
SPI_MOSI	40	GPIO4	SPI2_MOSI	UART2_TX	-	Y	boot_config[3]

SIM7090G HW Peripheral



Pin Name	Pin No.	Mode 1	Mode 2	Mode 3	Mode 4	Interrupt	Remark
W_DISABLE	41	GPIO27	-	-	-	Y	
UART_TXD1	60	GPIO0	UART1_TX	SPI1_MOSI	-	Y	QAPI_UART_PORT_001_E
UART_RXD1	61	GPIO1	UART1_RX	SPI1_MISO	-	Y	QAPI_UART_PORT_001_E
SPI_CS	63	GPIO6	SPI2_CS	UART2_CTS	I2C2_SDA	Y	
UART_RI	76	GPIO26	-	-	-	Y	
AP_READY	77	GPIO52	-	-	-	Y	
STATUS	78	GPIO10	SPI3_CS	I2C3_SDA	UART3_CTS	N	
NETLIGHT	79	GPIO11	SPI3_CLK	I2C3_SCL	UART3_RTS	N	
UART_DCD	90	GPIO28	-	-	-	Y	

Send and Receive AT Command

MCU

Send AT Command



Receive AT Response
return value



DAM

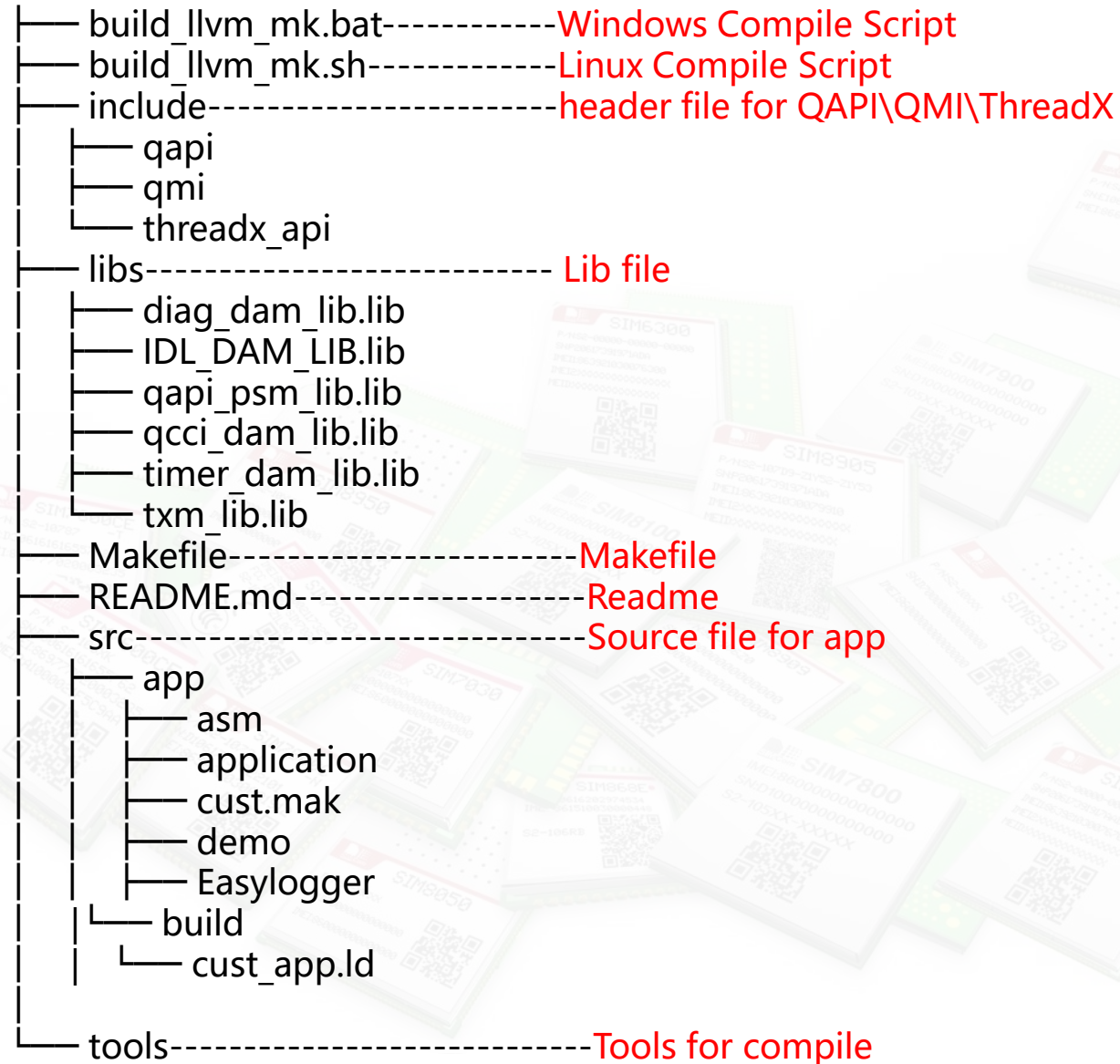
Send AT Command



Receive AT Response
return value



DAM SDK Structure



Note:



1. The entrance for DAM is **qcli_dam_app_start** in `ciot_sdk_9205\src\app\application\pal_module.c`, developer could create new task and app from this entrance.
2. DAM is based on ThreadX and QAPI, developer could all APIs for them (in `ciot_sdk_9205\include`). ThreadX is used to create and manage new thread , QAPI is used to process HW and software peripheral such as UART\SPI\ADC\AT core\timer\IIC\USB and so on. Please refer to SIM70X0 Series_ThreadX_API_V1.00 and SIM70X0 Series_QAPI_V1.00.
3. SIMCom suggest for modem operation(such as TCP\UDP\MQTT\SSL\FTTP\FTP.etc), please use AT cmd according to related application note instead of using QAPI.
4. If developer need to add other source file folder and source file, need to add include patch in `ciot_sdk_9205\src\app\cust.mak`

###

```
CUST_SRC_PATH += ./src/app/atfwd ./src/app/location ./src/app/net ./src/app/qcli \  
./src/app/simcom ./src/xxxxx
```

DAM SDK Compile Process

To compile SDK, we need:

- **LLVM 4.0.11(Snapdragon-llvm-4.0.11-windows64)**
- **python2.7.15 (python-2.7.15.amd64)**
- **ciot_sdk_9205**

DAM SDK Compile Process



Step 1: Unzip Snapdragon-llvm-4.0.11-windows64 on WIN64 OS(suggest WIN10),for example

in D:\SupportedCase\EAT\SIM7070G

Step 2: Install python-2.7.15.amd64(default setting) to C:\Python27

Step 3: Open ciot_sdk_9205\tools\opt.mak,modify path for LLVM and Python27

```
ifeq ($(OS), WINDOWS)
LLVMTOOLCHAIN_PATH:=D:\SupportedCase\EAT\SIM7070G\Snapdragon-llvm-4.0.11-windows64\bin
LLVMTOOLCHAIN_PATH_STANDARDS:=D:\SupportedCase\EAT\SIM7070G\Snapdragon-llvm-4.0.11-
windows64\armv7m-none-eabi\libc\include
LLVMLIB:=D:\SupportedCase\EAT\SIM7070G\Snapdragon-llvm-4.0.11-windows64\lib\clang\4.0.11\lib
LLVMLINK_PATH:=D:\SupportedCase\EAT\SIM7070G\Snapdragon-llvm-4.0.11-windows64\tools\bin
LLVMTOOLCHAIN:=clang.exe
LLVMLINKTOOL:=clang++.exe
PYTHON_PATH := c:/Python27/python.exe
```


DAM SDK Compile Process



Step 4: Open CMD line in Windows, then run build_llvm_mk.bat ,in iot_sdk_9205\bin\ there will be app file generated(cust_app.bin)

```
D:\SupportedCase\EAT\SIM7070G\ciot_sdk_9205>build_llvm_mk.bat  
"CC src/app/atfwd/atfwd_cli_app_module.c"
```

.....

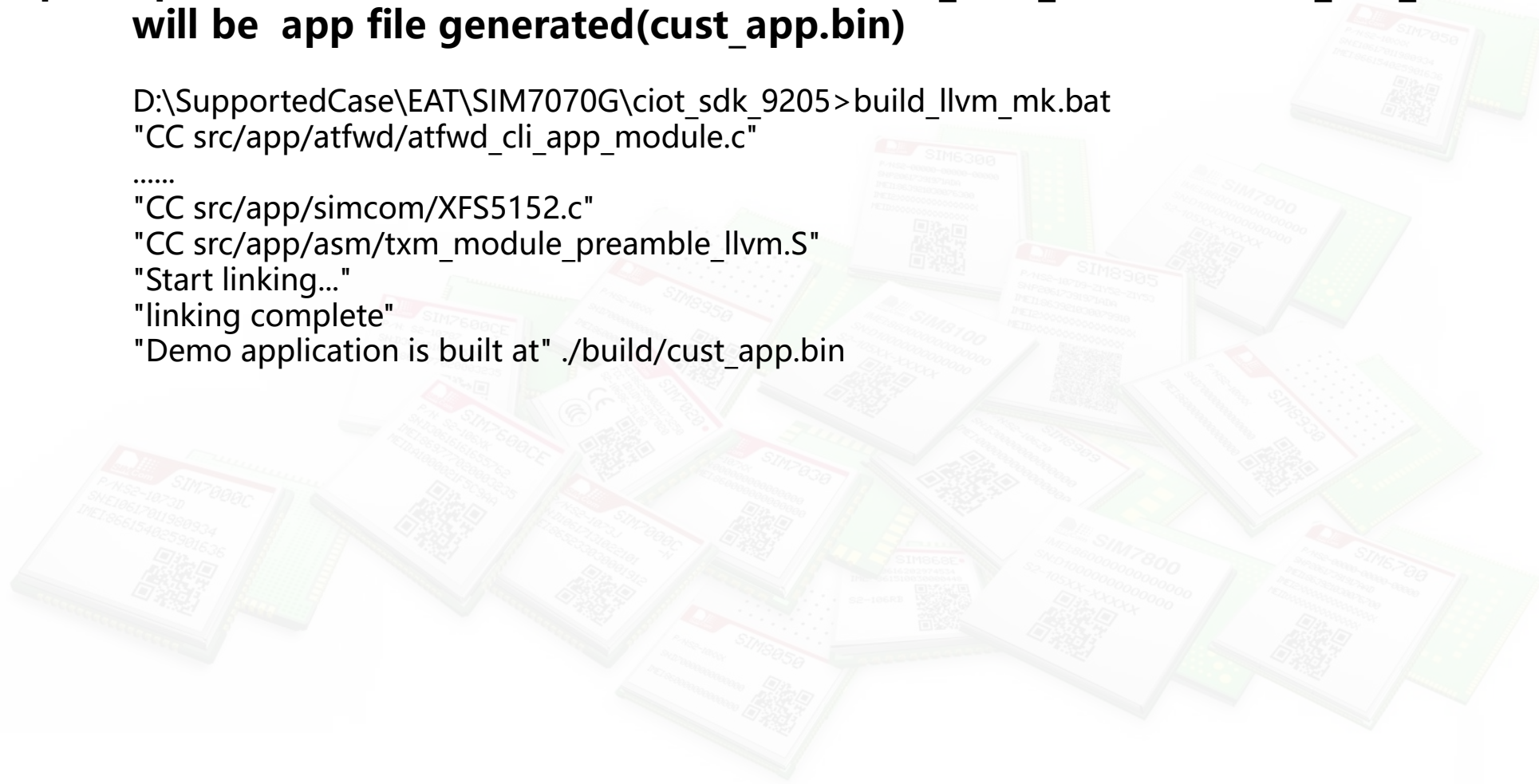
```
"CC src/app/simcom/XFS5152.c"
```

```
"CC src/app/asm/txm_module_preamble_llvm.S"
```

```
"Start linking..."
```

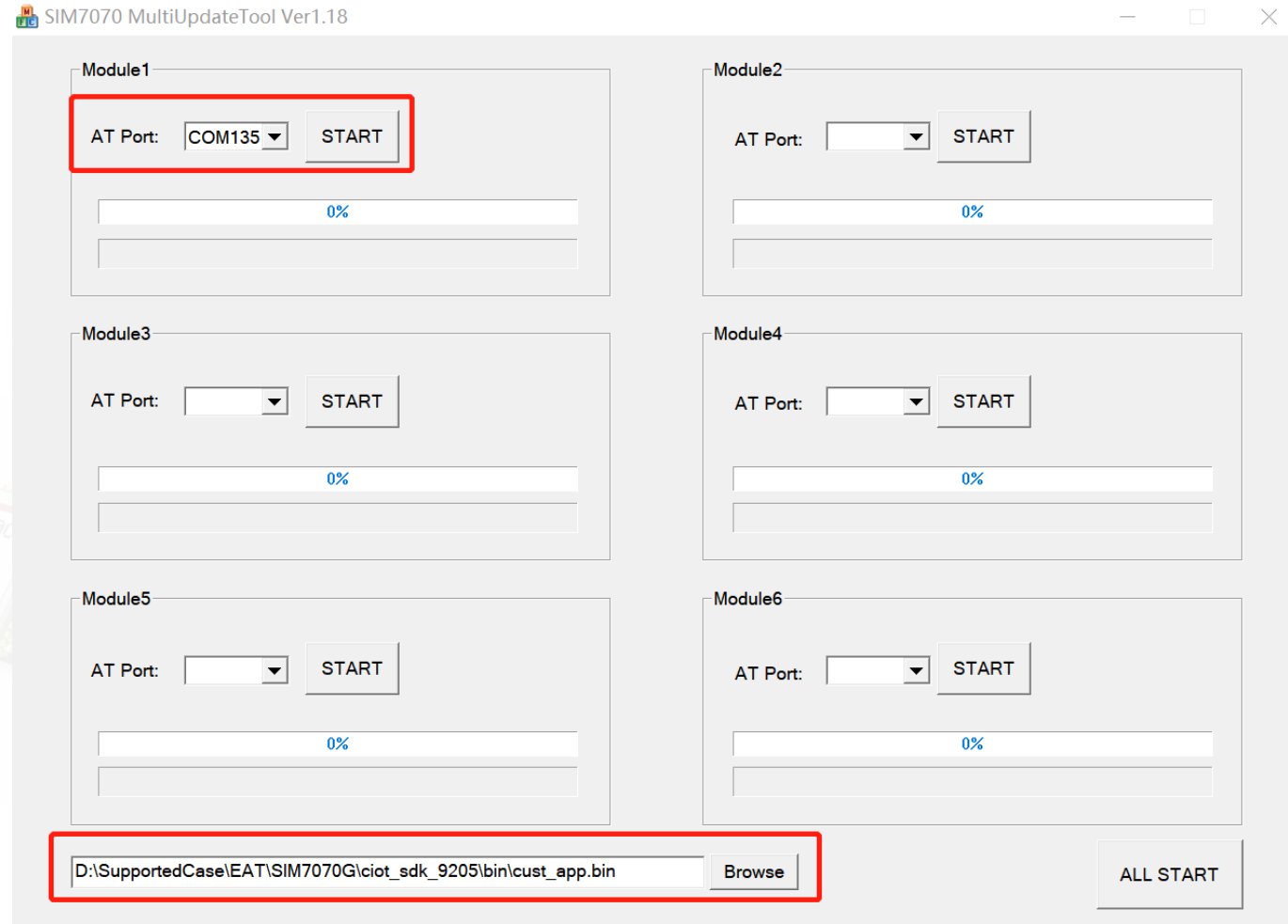
```
"linking complete"
```

```
"Demo application is built at" ./build/cust_app.bin
```



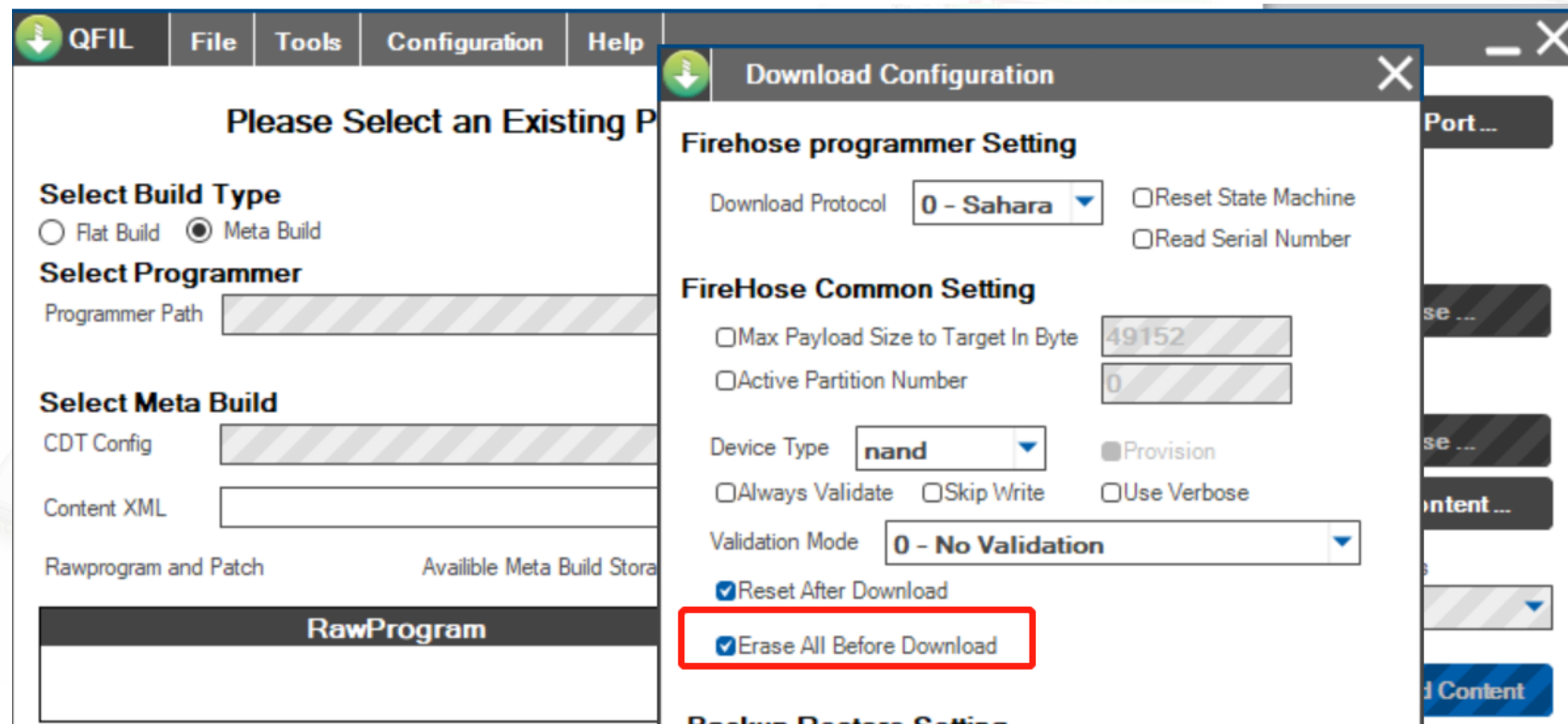
DAM Download Process

Open SIM7070 MultiUpdateTool Ver1.18.exe, choose AT port(it can be UART or USB), browse cust_app.bin file, power up module and click Start,after reboot app will start run.



DAM APP Erase Process

If need to erease app, need to upgrade FW for module with QFIL tool from QPST with “Erase All Before Download” ,then download FW and qmbn file ,about QPST tool please ask SIMCom FAE for help.



Thanks!

Modules for a Connected World

www.simcomm2m.com

SUNSEA IOT

SIMCom Wireless Solutions Co.,Ltd. ©All rights reserved

Add: Building B, SIM Technology Building, No.633 Jinzhong Road,
Changning District, Shanghai P.R.China 200335

Tel: +86 21 31575100/ +86 21 31575200

Email: simcom@simcom.com

Web: <http://www.simcomm2m.com>

SUNSEA IOT